



Kubernetes

This page contains a list of commonly used kubectl commands and flags.

Viewing and finding resources

Nodes

```
kubectl get no # Display all node information
kubectl get no -o wide # Show more information about all nodes
kubectl describe no # Display node details
kubectl get no -o yaml # Display node details in yaml format
kubectl get node --selector=[label_name] # Filter the node with the specified label
kubectl get nodes -o jsonpath='{.items[*].status.addresses[?(@.type="ExternalIP")]'
# Output the field information defined by the jsonpath expression
kubectl top node [node_name] # Display node (CPU/memory/storage) usage
```

Resource name: nodes, abbreviation: no

Pods

```
kubectl get po # Display all container group information
kubectl get po -o wide
kubectl describe po
kubectl get po --show-labels # View the labels of the container group
kubectl get po -l app=nginx
kubectl get po -o yaml
kubectl get pod [pod_name] -o yaml --export
kubectl get pod [pod_name] -o yaml --export > nameoffile.yaml
# Export container group information to yaml file in yaml format
kubectl get pods --field-selector status.phase=Running
# Use the field selector to filter out container group information
```

Resource name: pods, abbreviation: po

Namespaces

```
kubectl get ns
kubectl get ns -o yaml
kubectl describe ns
```

Resource name: namespaces, abbreviation: ns

Deployments

```
kubectl get deploy
kubectl describe deploy
kubectl get deploy -o wide
kubectl get deploy -o yaml
```

Resource name: deployments, abbreviation: deploy

Services

```
kubectl get svc
kubectl describe svc
kubectl get svc -o wide
kubectl get svc -o yaml
kubectl get svc --show-labels
```

Resource name: services, abbreviation: svc

Daemon Sets

```
kubectl get ds
kubectl describe ds --all-namespaces
kubectl describe ds [daemonset_name] -n [namespace_name]
kubectl get ds [ds_name] -n [ns_name] -o yaml
```

Resource name: daemonsets, abbreviation: ds

Events

```
kubectl get events
kubectl get events -n kube-system
kubectl get events -w
```

Resource name: events, abbreviation: ev

Logs

```
kubectl logs [pod_name]
kubectl logs --since=1h [pod_name]
kubectl logs --tail=20 [pod_name]
kubectl logs -f -c [container_name] [pod_name]
kubectl logs [pod_name] > pod.log
```

Service Accounts

```
kubectl get sa
kubectl get sa -o yaml
kubectl get serviceaccounts default -o yaml > ./sa.yaml
kubectl replace serviceaccount default -f ./sa.yaml
```

Resource name: serviceaccounts, abbreviation: ev

Replica Sets

```
kubectl get rs
kubectl describe rs
kubectl get rs -o wide
kubectl get rs -o yaml
```

Resource name: replicaset, abbreviation: rs

Roles

```
kubectl get roles --all-namespaces
kubectl get roles --all-namespaces -o yaml
```

Secrets

```
kubectl get secrets
kubectl get secrets --all-namespaces
kubectl get secrets -o yaml
```

Config maps

Resource name: configmaps, abbreviation: cm

```
kubectl get cm
```

```
kubectl get cm --all-namespaces
kubectl get cm --all-namespaces -o yaml
```

Ingresses

Resource name: ingresses, abbreviation: ing

```
kubectl get ing
kubectl get ing --all-namespaces
```

Persistent Volumes

Resource name: persistentvolumes, abbreviation: pv

```
kubectl get pv
kubectl describe pv
```

Persistent volume declaration

Resource name: persistentvolumeclaims, abbreviation: pvc

```
kubectl get pvc
kubectl describe pvc
```

storage class

Resource name: storageclasses, Abbreviation: sc

```
kubectl get sc
kubectl get sc -o yaml
```

Multiple resources

```
kubectl get svc, po
kubectl get deploy, no
kubectl get all
kubectl get all --all-namespaces
```

Updating resources

Taint

```
kubectl taint [node_name] [taint_name]
```

Label

```
kubectl label [node_name] disktype=ssd  
kubectl label [pod_name] env=prod
```

Maintain/Schedulable

```
kubectl cordon [node_name] # node maintenance  
kubectl uncordon [node_name] # node is schedulable
```

clear

```
kubectl drain [node_name] # empty the node
```

Node/Pod

```
kubectl delete node [node_name]  
kubectl delete pod [pod_name]  
kubectl edit node [node_name]  
kubectl edit pod [pod_name]
```

Stateless/Namespaced

```
kubectl edit deploy [deploy_name]  
kubectl delete deploy [deploy_name]  
kubectl expose deploy [deploy_name] --port=80 --type=NodePort  
kubectl scale deploy [deploy_name] --replicas=5  
kubectl delete ns  
kubectl edit ns [ns_name]
```

Service

```
kubectl edit svc [svc_name]  
kubectl delete svc [svc_name]
```

Daemon set

```
kubectl edit ds [ds_name] -n kube-system  
kubectl delete ds [ds_name]
```

```
kubectl edit sa [sa_name]
kubectl delete sa [sa_name]
```

```
kubectl annotate po [pod_name] [annotation]
kubectl annotateno [node_name]
```

Create resources

```
kubectl create -f [name_of_file]
kubectl apply -f [name_of_file]
kubectl run [pod_name] --image=nginx --restart=Never
kubectl run [pod_name] --generator=run-pod/v1 --image=nginx
kubectl run [pod_name] --image=nginx --restart=Never
```

```
kubectl create svc nodeport [svc_name] --tcp=8080:80
```

```
kubectl create -f [name_of_file]
kubectl apply -f [name_of_file]
kubectl create deploy [deploy_name] --image=nginx
```

```
kubectl run [pod_name] --image=busybox --rm -it --restart=Never --sh
```

```
kubectl create deploy [deploy_name] --image=nginx --dry-run -o yaml > deploy.yaml
kubectl get po [pod_name] -o yaml --export > pod.yaml
```

```
kubectl -h
kubectl create -h
kubectl run -h
kubectl explain deploy.spec
```

Miscellaneous

APIs

```
kubectl get --raw /apis/metrics.k8s.io/
```

Information

```
kubectl config
kubectl cluster-info
kubectl get componentstatus
```

Also See

[Kubernetes Official Documentation \(kubernetes.io\)](https://kubernetes.io)

Related Cheatsheet

ES6 Cheatsheet

Quick Reference

Express Cheatsheet

Quick Reference

JSON Cheatsheet

Quick Reference

TOML Cheatsheet

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